

MRINMOY KUMAR MAITY

Junior AI/ML Engineer — Generative AI Python Developer

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SUMMARY

Entry-level AI/ML Engineer with hands-on experience building Python applications using LLMs, RAG, LangChain, LangGraph, Pydantic, Flask, and FastAPI. Developed resume screening and semantic search systems with structured LLM outputs, vector retrieval, local embeddings, and REST API integration. Strong foundation in Python, SQL, machine learning, and data analysis, with a focus on building reliable and practical AI solutions.

SKILLS

Programming: Python, SQL

AI/ML: Machine Learning, Generative AI, LLMs, Prompt Engineering, RAG, AI Agents, Vector Databases, Semantic Search, Ollama

Libraries: LangChain, LangGraph, Scikit-learn, Pandas, NumPy, Pydantic, Flask

Analytics: Power BI, Excel, Matplotlib, EDA, Data Cleaning

Tools/Frameworks: FastAPI, Streamlit, REST APIs, Git, GitHub, MySQL

EXPERIENCE

Deloitte Australia Data Analytics Virtual Experience Program

May 2025 – Jul 2025

Forage – Remote

- Analyzed business and forensic datasets to identify patterns, inconsistencies, and actionable insights.
- Developed an interactive Tableau dashboard to visualize key findings and support business decisions.
- Used Excel to classify data, perform analysis, and communicate conclusions through a structured business report.

PROJECTS

Deep Packet Inspection and Network Traffic Analyzer

Aug 2026 – Present

Python, TCP/IP, PCAP, TLS, HTTP, DNS, Wireshark, JSON, Git [GitHub](#)

- Reimplemented and enhanced a C++ Deep Packet Inspection engine in Python to analyze offline PCAP files containing Ethernet, VLAN, IPv4, TCP, and UDP packets.
- Extracted domains using TLS SNI, HTTP Host headers, and DNS queries; implemented bidirectional flow tracking, application classification, and IP-, application-, and domain-based traffic filtering.
- Generated filtered PCAP files and structured JSON reports with robust error handling and automated tests; successfully analyzed a 77-packet test capture with zero parsing errors.

Candidate AI Representative

Aug 2026

Python, FastAPI, React, Groq API, Pydantic, Vite

- Built and deployed a full-stack, resume-grounded AI assistant that provides streaming answers using the candidate's active resume and structured profile data.
- Implemented PDF resume upload, download and text extraction, resume-based interview question generation, job-description suitability scoring (0–100), and a “Why Hire This Candidate?” mode.
- Added speech-to-text, text-to-speech, multilingual support for English, Hindi and Bengali, conversation history, and PDF chat export.

Deployed the frontend on Vercel and the FastAPI backend on Render. [Live Demo](#) — [GitHub](#)

AI-Powered Vector Database & RAG System

Jul 2026 – Present

Python, Flask, Ollama, Llama 3.2, RAG, REST API [GitHub](#)

- Developed a Python-based semantic search and RAG system with document chunking, vector retrieval, and interactive visualization.
- Integrated Ollama's nomic-embed-text and Llama 3.2 for local document embeddings and context-aware question answering.
- Built Flask REST APIs for vector search, document management, semantic retrieval, and algorithm benchmarking.

ResumeParser AI – Intelligent Candidate Screening System

Python, Groq LLM, Pydantic [GitHub](#)

- Built an AI-powered resume screening system that parses PDF/DOCX resumes and extracts skills, education, and experience.
- Generated AI-based candidate-job match scores with structured JSON output, ranking, and missing-skill identification.

EDUCATION

Bachelor of Computer Applications (BCA)

Expected Nov 2027

Kharagpur College, West Bengal — Specialization: Computer Science

Higher Secondary Education (Class XII)

2023

Jhargram Kumud Kumari Institution, West Bengal — Computer Science

CERTIFICATIONS

- Introduction to Agent Skills & Claude 101 – Anthropic (Jul 2026)
- Make Agentic AI Work for You – IBM (Jun 2026) | Data Analysis Using Python – IBM (Aug 2025)
- GenAI in Data Analytics – Meta (Jul 2025) | Google for Education Certifications – Google (Oct 2025)